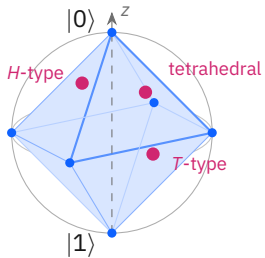
Preparing a logical magic state on a noisy quantum computer

Parity-check detection, dynamical decoupling and readout correction on IBM Aachen

Qompute 2026 · Track 3: Can We Trust a Noisy Quantum Computer?

Runzhe Mo, Mingyu Li

What is a magic state, and why does it matter?



Blue: stabilizer states. Red: magic-state examples.

Preparing one physical magic state

$|0\rangle$ — $R_y(\theta)$ — $R_z(\phi)$ — $|M\rangle$

$\theta = \arccos(1/\sqrt{3})$, $\phi = \pi/4$: the tetrahedral state

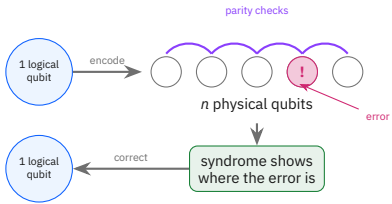
Bravyi & Kitaev, Phys. Rev. A 71, 022316 (2005); Reiher et al., PNAS 114, 7555 (2017).

Clifford gates (H, S, CNOT) acting on stabilizer states can be simulated on a classical computer.

Adding **magic states** gives a universal set of quantum gates.

In fault-tolerant machines each T gate consumes one magic state, so its quality limits the whole computation (e.g. chemistry simulation).

Quantum error correction: redundancy plus parity checks



Example: the [[5, 1, 3]] code

	q_1	q_2	q_3	q_4	q_5
g_1	X	Z	Z	X	I
g_2	I	X	Z	Z	X
g_3	X	I	X	Z	Z
g_4	Z	X	I	X	Z

$$|0_L\rangle = \frac{1}{4}\sum_g g\,|00000\rangle$$

$$= \tfrac{1}{4}\left(|00000\rangle + |00101\rangle + |01001\rangle + |01010\rangle + \cdots - |00011\rangle - |00110\rangle - \cdots\right)$$

Spread one **logical qubit** over many physical qubits, so a single fault does not destroy it.

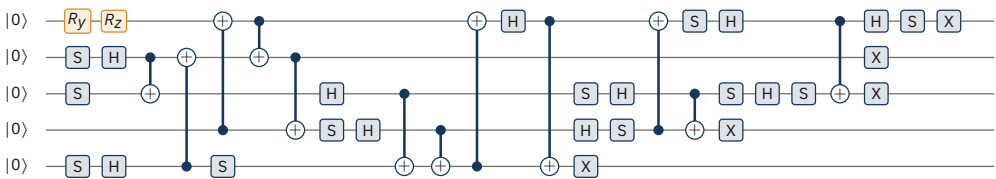
Measure **parity checks** (stabilizers): they reveal an error without reading the data.

A code $[[n, k, d]]$ has n physical qubits, k logical qubits and distance d; $d = 3$ corrects any single-qubit error.

Laflamme, Miquel, Paz & Zurek, Phys. Rev. Lett. 77, 198 (1996).

Method A: prepare one magic state and encode it, no check

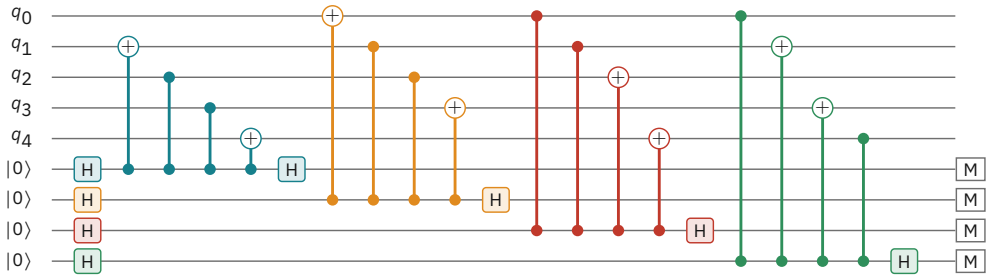
Prepare $|T\rangle$ on q_0, then apply the Clifford encoder to $q_0 \dots q_4$



$R_y(\theta)$, $R_z(\pi/4)$ with $\theta = \arccos(1/\sqrt{3})$; 12 two-qubit gates in the encoder. The output is $|T_L\rangle$ on $q_0 \dots q_4$.

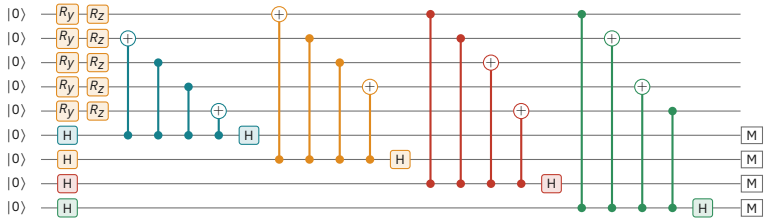
Method A, continued: check the syndrome and keep only $s = 0000$

Input: the encoded state from the previous page on $q_0 \ldots q_4$; fresh ancillas $q_5 \ldots q_8$



16 two-qubit gates in the checks. Checks: g_1 teal, g_2 amber, g_3 red, g_4 green.

Method B: prepare five magic states, then project with the checks



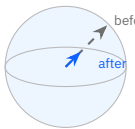
Inputs: $R_y(\pi - \theta)$ then $R_z(5\pi/4)$ on each of $q_0 \ldots q_4$ gives $|\bar{T}\rangle$, the antipodal state of $|T\rangle$.

Projection: keep only $s = 0000$. The surviving state is $|T_L\rangle$; ideal yield $1/6$, 16 two-qubit gates, 5 magic states per attempt.

Bravyi & Kitaev, Phys. Rev. A 71, 022316 (2005).

Three kinds of noise in the model

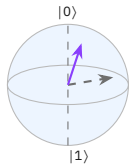
Gate error



$\rho \to (1-p)\,\rho + p\,\frac{I}{2^n}$

depolarizing, calibrated per gate, plus relaxation during the gate

Idle decay T_1, T_2

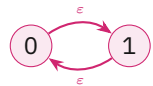


$\rho_{11} \to e^{-t/T_1}\rho_{11}$

$\rho_{01} \to e^{-t/T_2}\rho_{01}$

acts on every waiting qubit, also during the readout wait

Readout error



$P(\text{rec} \neq \text{true}) = \varepsilon$

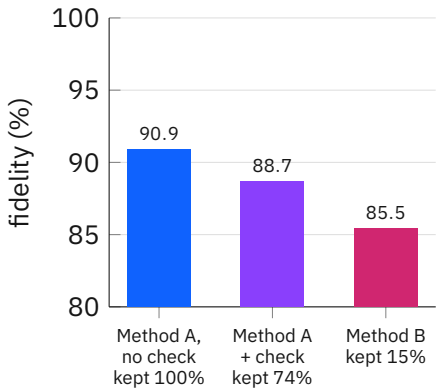
on the syndrome ancillas: false reject or false accept

IBM Quantum Documentation, “Build noise models”,

`https://quantum.cloud.ibm.com/docs/en/guides/build-noise-models`.

Noise simulation: the check lowers the delivered fidelity

Delivered fidelity F, [[5, 1, 3]] code



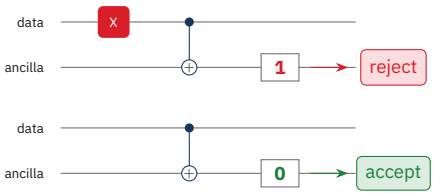
Where it sits: $F = P_{\text{code}} \times F_{\text{in code}}$

	F	P_{code}	$F_{\text{in code}}$
Method A, no check	90.9%	91.3%	99.5%
Method A + check	88.7%	89.5%	99.1%
Method B	85.5%	85.7%	99.7%

P_{code}: population inside the code; $F_{\text{in code}}$: fidelity within the code

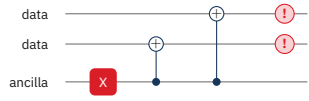
What the syndrome check catches, and what it brings along

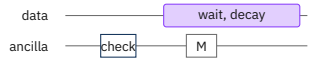
Caught: data errors before the check



Any single *X*, *Y* or *Z* flips at least one check.

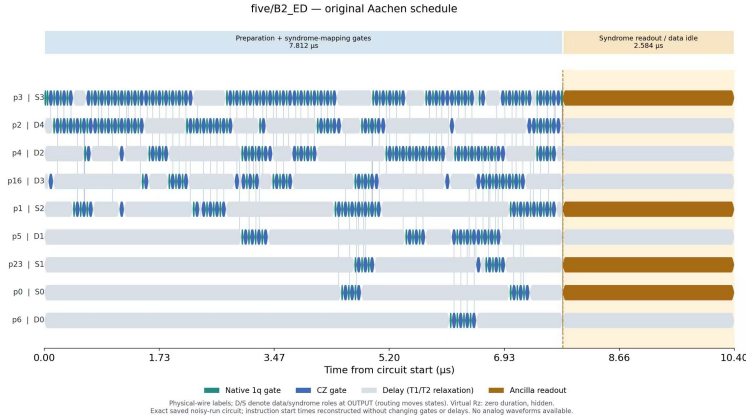
Brought along, or missed







The syndrome readout wait is where the fidelity is lost

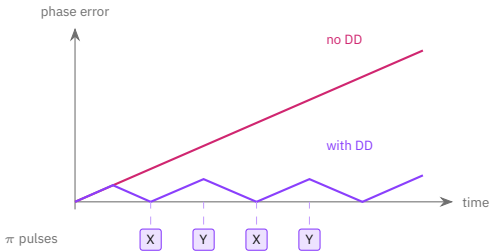


Switching off relaxation in the 2.58 µs readout wait alone: fidelity 88.7% $\rightarrow$ 93.1% (simulation).

IBM Quantum Documentation, “Build noise models”.

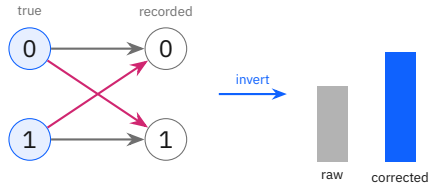
Two mitigation tools: dynamical decoupling and readout correction

Dynamical decoupling (XY4)



slow phase drift is flipped back and forth and cancels

Readout correction



measured flips are undone in the estimate; the state itself is unchanged

IBM Quantum Documentation, “Error suppression”; “Readout error mitigation with Sampler”.

IBM Aachen hardware: the two mitigations and their combination

Mitigation	acts on	fidelity F	95% interval	runs kept
none	–	60.6%	[59.5, 61.7]	37.7%
readout correction	estimate	68.4%	[67.1, 69.6]	37.7%
XY4 DD	prepared state	69.9%	[68.9, 70.9]	38.6%
XY4 DD + readout correction	state and estimate	79.1%	[78.0, 80.3]	38.6%

$[[5, 1, 3]]$, tetrahedral state, Method A + check, 2,048 shots per setting. Runs kept use the dedicated acceptance circuit; the XY4 and no-DD jobs ran in sequence.

Thank you for listening

Questions are welcome